



Thriviving Scholars
SAT PREPARATION STUDIO

2026 • READ + PRINT EDITION

 **DIGITAL SAT-STYLE PRACTICE**

SAT Diagnostic Test 6

A clean, linked solution book for reviewing every question, checking the answer key, and understanding the reasoning behind each choice.

Solution Book

Read-only review edition with all questions, answers, and worked solutions.
Independent practice material; not an official College Board publication.

98

QUESTIONS

4

MODULES

1

COMPLETE BOOK

Review the test in Bluebook order.

The source test is organized into the same four locked modules used by a digital SAT-style session. This edition keeps that framing, then adds a linked answer key and one worked solution for every item.

Reading & Writing

Modules 1–2 · 27 questions each · 32 minutes each

Math

Modules 1–2 · 22 questions each · 35 minutes each

Response formats

87 multiple-choice items · 11 student-produced responses

- ◆ **Read-only solution edition.** Correct answers are not highlighted in the question pages; use the answer key or jump directly to a worked solution. Math is typeset with MathJax for clean screen and print review.

Math reference: The volume of a cone is $V = \frac{1}{3}\pi r^2 h$.

CONTENTS

Four modules. One review path.



Reading & Writing · Module 1

27 questions · 32 min · no calculator



Reading & Writing · Module 2

27 questions · 32 min · no calculator



Math · Module 1

22 questions · 35 min · calculator allowed



Math · Module 2

22 questions · 35 min · calculator allowed



PART I

Questions

98 source questions · answer choices preserved



SAT DIAGNOSTIC TEST 6

Reading & Writing · Module 1

27 questions

32 min

No calculator

Questions for the Reading & Writing · Module 1 section. Craft and Structure, Information and Ideas, Standard English Conventions, and Expression of Ideas.

Question 1

Reading & Writing · Module 1 · Diagnostic Test 6

Craft and Structure

Multiple choice

An economist predicted that a congestion charge would sharply reduce weekday driving. After the city returned much of the revenue to residents through transit credits, however, the decline in driving was smaller than expected. The credits appear to have _____ the charge's effect on travel behavior without eliminating it.

Which choice completes the text with the most logical and precise word or phrase?

A replicated

B obscured

C tempered

D reversed

Reading & Writing · Module 1 · Question 1

Worked solution

Answer C

VOCABULARY IN CONTEXT

- 1 The credits made the reduction in driving smaller, so they moderated the charge's effect without eliminating it.
- 2 "Tempered" means moderated or softened. It is the only choice that matches the relationship described.

Reasoning check The effect became weaker, not hidden, copied, or reversed.

Question 2

Reading & Writing · Module 1 · Diagnostic Test 6

Craft and Structure

Multiple choice

Because only a small portion of the settlement's records survived, historian Lena Ortiz treats her reconstruction of its trade network as _____ rather than definitive; newly discovered ledgers could substantially alter the picture.

Which choice completes the text with the most logical and precise word or phrase?

- ☐ A exhaustive
- ☐ B redundant
- ☐ C provisional
- ☐ D incontrovertible

Reading & Writing · Module 1 · Question 2

Worked solution

Answer C

INFERENCE FROM EVIDENCE

- 1 The passage says that newly found ledgers could change the accepted account of the settlement's trade network.
- 2 An account that may be revised when more records are found is provisional.

Reasoning check The evidence supports a cautious conclusion rather than a definitive reconstruction.

Question 3

Reading & Writing · Module 1 · Diagnostic Test 6

Craft and Structure

Multiple choice

Researchers observed that a desert moss recovered more quickly after brief rain when neighboring moss patches were present. One explanation is that neighboring patches simply mark locations where the soil already retains more water. Yet when the researchers moved equal amounts of soil among plots, recovery remained faster beside living patches, suggesting that the neighbors themselves contribute to the effect.

Which choice best describes the function of the underlined sentence in the text as a whole?

- ☐ A It identifies a flaw in the experimental method that prevents the researchers from drawing a conclusion.
- ☐ B It explains why the researchers expected moved soil to improve moss recovery.
- ☒ C It presents an alternative explanation that the subsequent evidence helps rule out.
- ☐ D It states the researchers' main conclusion before the evidence for that conclusion is introduced.

Reading & Writing · Module 1 · Question 3

Worked solution

Answer C

FUNCTION OF A SENTENCE

- 1 The underlined sentence offers retained soil moisture as an alternative explanation for the faster recovery.
- 2 The next evidence—that soil was moved beside living patches—undercuts that alternative and supports the researchers' interpretation.

Reasoning check The sentence introduces a possible explanation that the surrounding evidence helps rule out.

Question 4

Reading & Writing · Module 1 · Diagnostic Test 6

Craft and Structure

Multiple choice

Text 1

Planting trees can reduce summer temperatures in cities, but the cooling benefit depends heavily on canopy size. Programs that maximize the number of newly planted trees may therefore produce disappointing results if many trees remain small.

Text 2

In a five-year trial, drought-tolerant native trees initially produced less shade than several fast-growing species. By year five, however, the native trees had much higher survival rates, and their total surviving canopy area was greater.

Based on the texts, both authors would most likely agree with which statement?

- ☐ A Native trees provide greater shade than other species immediately after planting.
- ☐ B The long-term cooling value of a planting program depends on more than the number of trees initially planted.
- ☐ C Fast-growing tree species should never be used in urban planting programs.
- ☐ D Tree survival matters only when cities are unable to plant large numbers of trees.

Reading & Writing · Module 1 · Question 4

Worked solution

Answer B

PAIRED-TEXT COMPARISON

- 1 Both passages show that the long-term value of an urban tree-planting program depends on more than the initial number of trees.
- 2 Neither passage claims that simply increasing the initial number of trees guarantees the same result.

Reasoning check The shared focus is the condition and duration of cooling, not the initial tree count alone.

Question 5

Reading & Writing · Module 1 · Diagnostic Test 6

Craft and Structure

Multiple choice

When two instruments produce nearly identical signals, the new sensor can _____ them only if the signals differ by more than its measurement uncertainty. Below that threshold, the sensor records the signals as effectively indistinguishable.

Which choice completes the text with the most logical and precise word or phrase?

A substitute for

B differentiate between

C reconcile

D amplify

Reading & Writing · Module 1 · Question 5

Worked solution

Answer B

VOCABULARY IN CONTEXT

1 The sensor is useful only when it can tell one signal apart from another despite background noise.

2 “Differentiate between” means distinguish, which describes that function precisely.

Reasoning check The sentence requires a verb phrase about distinguishing signals.

Question 6

Reading & Writing · Module 1 · Diagnostic Test 6

Craft and Structure

Multiple choice

Biologists found that a certain beetle remains active at temperatures that cause related species to become sluggish. The beetle's cell membranes contain unusually high proportions of lipids that remain flexible in cold conditions. This flexibility allows proteins embedded in the membranes to continue functioning as temperatures fall. The finding offers a physiological explanation for the beetle's cold-weather activity.

Which choice best describes the function of the underlined sentence in the text as a whole?

- ☐ A It questions whether membrane composition differs among beetle species.
- ☐ B It supplies a mechanism connecting a cellular trait to the observed behavior.
- ☐ C It introduces a competing explanation for the beetle's activity.
- ☐ D It describes the experimental procedure used to measure beetle activity.

Reading & Writing · Module 1 · Question 6

Worked solution

Answer **B**

FUNCTION AND MECHANISM

- 1 The passage links the organism's flexible lipids to the continued operation of membrane proteins in cold conditions.
- 2 That mechanism explains why the organism can remain active when the temperature falls.

Reasoning check The selected idea connects a cellular property directly to cold-weather activity.

Question 7

Reading & Writing · Module 1 · Diagnostic Test 6

Craft and Structure

Multiple choice

Text 1

A translator of poetry should preserve unusual syntax whenever possible. Even if the resulting English feels unfamiliar, the strangeness may be part of the poem's artistic effect.

Text 2

Reproducing a source poem's syntax word for word can sometimes distort its effect because grammatical structures carry different associations across languages. A translation may need different syntax to create a comparable experience for the new reader.

How would the author of Text 2 most likely respond to the main claim in Text 1?

- ☐ A By claiming that syntax has little influence on a poem's artistic effect.
- ☐ B By agreeing that unfamiliar syntax always produces the most faithful translation.
- ☐ C By arguing that preserving syntax is valuable only when doing so also preserves the source poem's effect.
- ☐ D By arguing that poetry cannot be meaningfully translated between languages.

Reading & Writing · Module 1 · Question 7

Worked solution

Answer C

PAIRED-TEXT COMPARISON

- 1 Text 1 values preserving the original syntax, while Text 2 adds an important qualification.
- 2 Text 2 accepts a change in syntax when the change is needed to preserve the original effect or meaning.

Reasoning check The second author's position is conditional: syntax matters when it carries the relevant effect.

Question 8

Reading & Writing · Module 1 · Diagnostic Test 6

Information and Ideas

Multiple choice

Many migratory bird species have begun arriving at northern breeding grounds earlier in spring. Yet the insects on which some birds feed have shifted their seasonal peaks by different amounts. Species whose arrival dates track insect peaks closely have maintained nesting success, whereas species whose schedules have changed less have experienced declines. The pattern suggests that earlier migration alone is not necessarily beneficial.

Which choice best states the main idea of the text?

- ☐ A The effect of earlier migration on nesting success depends on whether migration remains synchronized with food availability.
- ☐ B Bird species that arrive earliest always have the highest nesting success.
- ☐ C Insect populations have generally declined because birds now arrive earlier.
- ☐ D Most migratory birds are unable to alter the timing of their spring migration.

Reading & Writing · Module 1 · Question 8

Worked solution

Answer **A**

MAIN IDEA

- 1 The birds' migration is successful when it coincides with peaks in the insect population.
- 2 This timing supplies the food needed during the migration, so synchrony is the central point.

Reasoning check The passage emphasizes matching migration timing with food availability.

Question 9

Reading & Writing · Module 1 · Diagnostic Test 6

Information and Ideas

Multiple choice

A materials scientist tested a coating intended to reduce corrosion. Identical metal samples were placed in saltwater; half were coated and half were not. After four weeks, coated samples had lost less mass on average. The scientist repeated the trial using fresh samples and obtained a similar difference.

Which conclusion is best supported by the text?

- ☐ A The second trial proves that no uncontrolled variable affected either experiment.
- ☐ B Under the tested conditions, the coating is associated with reduced corrosion, and the repeated trial strengthens confidence in that result.
- ☐ C Uncoated metal corrodes only because saltwater removes material from its surface.
- ☐ D The coating prevents corrosion completely in all saltwater environments.

Reading & Writing · Module 1 · Question 9

Worked solution

Answer **B**

EVIDENCE AND CONCLUSION

- 1 The coated samples corroded less than the uncoated samples in the first comparison.
- 2 A second controlled comparison produced the same pattern, strengthening the conclusion that the coating is associated with reduced corrosion.

Reasoning check Replication supports the stated association under the tested conditions; it does not prove every possible use.

Question 10

Reading & Writing · Module 1 · Diagnostic Test 6

Information and Ideas

Multiple choice

A researcher measured the electrical resistance of three materials before and after exposure to high humidity.

Material	Before (Ω)	After (Ω)
P	40	44
Q	75	90
R	120	126

Which choice best uses the data to support the claim that material Q is more sensitive to humidity than materials P and R?

- ☐ A Q's resistance after exposure was 90 Ω , which was lower than R's resistance after exposure.
- ☐ B Q's resistance increased by 20%, compared with 10% for P and 5% for R.
- ☐ C R had the greatest resistance both before and after exposure.
- ☐ D P's resistance increased by 4 Ω , while Q's increased by 15 Ω .

Reading & Writing · Module 1 · Question 10

Worked solution

Answer **B**

DATA INTERPRETATION

- 1 For P, the relative increase is $4 \div 40 = 10\%$.
- 2 For Q, it is $15 \div 75 = 20\%$; for R, it is $6 \div 120 = 5\%$.

Reasoning check Q has the greatest relative increase, even though its absolute increase is not the only quantity shown.

Question 11

Reading & Writing · Module 1 · Diagnostic Test 6

Information and Ideas

Multiple choice

Ecologist Mira Chen hypothesizes that diverse soil-microbe communities help grasslands recover from drought because different microbes perform overlapping nutrient-processing roles. If drought suppresses one microbial group, another may continue the same function.

Which finding, if true, would most directly support Chen's hypothesis?

- ☐ A Several microbial groups became less abundant during drought in nearly every plot.
- ☐ B Plots with greater microbial diversity maintained nutrient-cycling rates after drought even though the abundance of several microbial groups declined.
- ☐ C Some grass species grew taller in wet years than in dry years.
- ☐ D Plots with greater microbial diversity contained more plant species before drought began.

Reading & Writing · Module 1 · Question 11

Worked solution

Answer **B**

RESEARCH SUPPORT

- 1 The proposed redundancy mechanism predicts that several groups can perform overlapping nutrient-cycling roles.
- 2 The observation that nutrient cycling continued while some groups declined is direct evidence for that prediction.

Reasoning check The result matches functional overlap preserving an ecosystem process.

Question 12

Reading & Writing · Module 1 · Diagnostic Test 6

Information and Ideas

Multiple choice

After months of practicing alone, Nia finally played her composition for the ensemble. She kept her eyes on the score until the last chord faded. Then, hearing no response, she began gathering the pages. Only when the conductor quietly asked the musicians to play the piece again did Nia look up, startled.

Which choice best states what can reasonably be inferred from the text?

- ☐ A Nia preferred practicing alone to working with other musicians.
- ☐ B The conductor had previously told Nia that the composition would be performed twice.
- ☐ C The musicians had been unable to read the final chord of Nia's composition.
- ☐ D Nia had interpreted the ensemble's initial silence as a negative reaction.

Reading & Writing · Module 1 · Question 12

Worked solution

Answer **D**

INFERENCE FROM A CHARACTER'S RESPONSE

- 1 Nia first interprets the silence as a sign that the listeners are not interested.
- 2 When one listener asks her to play the recording again, she is surprised because the request conflicts with her initial negative interpretation.

Reasoning check Her reaction shows that she read the silence negatively before learning that the audience wanted more.

Question 13

Reading & Writing · Module 1 · Diagnostic Test 6

Information and Ideas

Multiple choice

A historian examining a port city's tax records notices that imports of timber rose sharply during years when local ship registrations also increased. The records do not indicate how the imported timber was used, and timber had many other uses in the city.

Which inference is best supported by the text?

- ☐ A Ship registrations increased because imported timber became less expensive.
- ☐ B Most imported timber must have been used for purposes unrelated to shipbuilding.
- ☐ C The increase in timber imports proves that every newly registered ship was built locally.
- ☐ D The records are consistent with the possibility that increased shipbuilding contributed to higher timber imports, but they do not establish that shipbuilding was the cause.

Reading & Writing · Module 1 · Question 13

Worked solution

Answer **D**

CAUSAL INFERENCE

- 1 The correlation is consistent with shipbuilding contributing to the increase in timber use.
- 2 However, timber may also have been used for housing, fuel, or other purposes, so the correlation alone cannot establish the cause.

Reasoning check A relationship in the data supports a possibility, not causal proof by itself.

Question 14

Reading & Writing · Module 1 · Diagnostic Test 6

Information and Ideas

Multiple choice

A battery was tested at four temperatures.

Temperature (°C)	Capacity after 100 cycles (%)
5	91
15	94
25	96
35	88

An engineer claims that, among the tested temperatures, increasing temperature improves capacity retention only up to a point.

Which choice best uses the data to support the engineer's claim?

- ☐ A The battery retains more than 85% capacity at every tested temperature.
- ☐ B Capacity retention differs by 3 percentage points between 5°C and 15°C.
- ☐ C The highest tested temperature is 35°C, and the lowest is 5°C.
- ☐ D Capacity retention rises from 91% at 5°C to 96% at 25°C but falls to 88% at 35°C.

Reading & Writing · Module 1 · Question 14

Worked solution

Answer D

DATA INTERPRETATION

- Retention rises from 91% to 96% between the first two periods.
- It then falls to 88%, so the overall change from the first to the last period is 3 percentage points lower.

Reasoning check The table shows an initial increase followed by a larger decrease.

Question 15

Reading & Writing · Module 1 · Diagnostic Test 6

Standard English Conventions

Multiple choice

Cloud cover had thickened throughout the afternoon, making navigation increasingly difficult. The expedition reached the plateau after sunset _____ the researchers postponed the survey until morning, when the markers would be easier to see.

Which choice completes the text so that it conforms to the conventions of Standard English?

A : therefore,

B ; therefore,

C , therefore,

D therefore

Reading & Writing · Module 1 · Question 15

Worked solution

Answer B

PUNCTUATION

- 1 The words before and after the blank are complete independent clauses.
- 2 A semicolon can join those clauses, and the conjunctive adverb “therefore” is followed by a comma.

Reasoning check The correct form is “; therefore,” because both clause boundaries are handled.

Question 16

Reading & Writing · Module 1 · Diagnostic Test 6

Standard English Conventions

Multiple choice

The glacier's boundary shifted unevenly between 2001 and 2021, with some sections retreating much faster than the others. Using satellite images collected over two decades, _____.

Which choice completes the text so that it conforms to the conventions of Standard English?

- ☐ A changes in the glacier's boundary were mapped by the researchers
- ☐ B a map of changes in the glacier's boundary was produced
- ☒ C the researchers mapped changes in the glacier's boundary
- ☐ D the glacier's boundary showed changes that the researchers mapped

Reading & Writing · Module 1 · Question 16

Worked solution

Answer C

MODIFIER PLACEMENT

- 1 “Using satellite images” is an introductory participial phrase.
- 2 The logical user of the images is the researchers, so placing the modifier with that subject avoids a dangling modifier.

Reasoning check The sentence must not make satellite images appear to be the researchers.

Question 17

Reading & Writing · Module 1 · Diagnostic Test 6

Standard English Conventions

Multiple choice

The insects were tested under red, blue, and white light as well as in darkness. A series of experiments conducted under different lighting conditions _____ that the insects orient primarily by contrast rather than by color.

Which choice completes the text so that it conforms to the conventions of Standard English?

A have shown

B show

C showing

D has shown

Reading & Writing · Module 1 · Question 17

Worked solution

Answer D

SUBJECT-VERB AGREEMENT

1 The subject is “a series,” not the plural noun “findings” inside the prepositional phrase.

2 Because “series” is singular here, the verb is “has shown.”

Reasoning check The main subject controls the singular verb.

Question 18

Reading & Writing · Module 1 · Diagnostic Test 6

Standard English Conventions

Multiple choice

Many documents in the collection are too fragile to withstand repeated handling. The museum digitized the collection so that researchers could examine _____ records without handling the fragile originals.

Which choice completes the text so that it conforms to the conventions of Standard English?

A they're

B their

C it's

D its

Reading & Writing · Module 1 · Question 18

Worked solution

Answer D

PRONOUN CASE AND POSSESSION

1 The collection belongs to the museum, so the blank needs a singular possessive pronoun.

2 “Its” supplies that possessive meaning; “it’s” would mean “it is.”

Reasoning check The sentence needs possession, not a contraction.

Question 19

Reading & Writing · Module 1 · Diagnostic Test 6

Standard English Conventions

Multiple choice

Weather at the remote site changes rapidly, and resupply can take several days. The field team needed three things before departure _____ updated maps, a calibrated sensor, and permission to enter the reserve.

Which choice completes the text so that it conforms to the conventions of Standard English?

A ,

B — and

C ;

D :

Reading & Writing · Module 1 · Question 19

Worked solution

Answer D

PUNCTUATION

- 1 The words before the blank form a complete clause and introduce the three items that follow.
- 2 A colon correctly introduces the list: “updated maps, a calibrated sensor, and permission to enter the reserve.”

Reasoning check The colon follows a complete clause and introduces the list.

Question 20

Reading & Writing · Module 1 · Diagnostic Test 6

Standard English Conventions

Multiple choice

The exhibition date had been fixed months in advance. By the time the public exhibition opened in June, conservators _____ every painting for structural damage and recorded the results in a shared database.

Which choice completes the text so that it conforms to the conventions of Standard English?

A are examining

B had examined

C will examine

D examine

Reading & Writing · Module 1 · Question 20

Worked solution

Answer **B**

SEQUENCE OF TENSES

1 The exhibition opened in June, and the conservators' examination occurred before that point.

2 The past perfect "had examined" marks the earlier of the two past actions.

Reasoning check The earlier past action takes the past-perfect form.

Question 21

Reading & Writing · Module 1 · Diagnostic Test 6

Standard English Conventions

Multiple choice

Researchers needed a tool that could identify weak transmitters in a noisy environment. The software was designed to measure signal strength, compare readings across sensors, and _____ the location of a source.

Which choice completes the text so that it conforms to the conventions of Standard English?

- ☐ A estimated
- ☐ B estimate
- ☐ C estimating
- ☐ D an estimate of

Reading & Writing · Module 1 · Question 21

Worked solution

Answer **B**

PARALLEL STRUCTURE

- 1 The verbs after “was designed to” should use the same base form.
- 2 “Measure,” “compare,” and “estimate” are parallel verbs, so B completes the series.

Reasoning check The three actions are expressed in parallel form.

Question 22

Reading & Writing · Module 1 · Diagnostic Test 6

Expression of Ideas

Multiple choice

The first prototype was inexpensive and easy to manufacture. _____ it failed after repeated exposure to heat, so the team replaced its outer casing before field testing.

Which choice completes the text with the most logical transition?

A However,

B For example,

C Therefore,

D Similarly,

Reading & Writing · Module 1 · Question 22

Worked solution

Answer **A**

TRANSITIONS

1 The first sentence says the prototype was inexpensive.

2 The next sentence presents an unexpected failure, so “However” signals the needed contrast.

Reasoning check The relationship is contrast, not addition or cause.

Question 23

Reading & Writing · Module 1 · Diagnostic Test 6

Expression of Ideas

Multiple choice

Early trials produced many duplicate readings because nearby sensors sampled the same events. The researchers increased the spacing between sensors, reducing overlap in their measurements. _____ the final data set contained fewer duplicate readings.

Which choice completes the text with the most logical transition?

A Consequently,

B For instance,

C Meanwhile,

D Nevertheless,

Reading & Writing · Module 1 · Question 23

Worked solution

Answer **A**

TRANSITIONS

- 1 Increasing the spacing between sensors reduced overlap in their measurements.
- 2 Fewer duplicate readings therefore followed, so “Consequently” correctly signals the result.

Reasoning check The second statement is an effect of the first.

Question 24

Reading & Writing · Module 1 · Diagnostic Test 6

Expression of Ideas

Multiple choice

While researching water purification, a student takes these notes:

- Method A removes dissolved salts by forcing water through a membrane.
- Method A requires substantial pressure and therefore considerable energy.
- Method B evaporates water and condenses the vapor.
- Method B can use waste heat from industrial processes.

The student wants to contrast the main energy demands of the two methods. Which choice most effectively uses relevant information from the notes?

- A** Both Method A and Method B can produce purified water from water containing dissolved salts.
- B** Industrial processes sometimes generate waste heat that can be used for other purposes.
- C** Method A uses a membrane, while Method B produces vapor that later condenses.
- D** Method A depends on energy-intensive pressure, whereas Method B can draw on waste heat for evaporation.

Reading & Writing · Module 1 · Question 24

Worked solution

Answer **D**

NOTES SYNTHESIS

- 1 Method A depends on pressure-intensive equipment, while Method B uses waste heat from an existing industrial process.
- 2 The notes therefore contrast the energy demands of the two methods.

Reasoning check The choice must preserve both the pressure requirement and the waste-heat advantage.

Question 25

Reading & Writing · Module 1 · Diagnostic Test 6

Expression of Ideas

Multiple choice

A student records these notes:

- A survey of 600 commuters found that 48% preferred expanded bus service.
- The survey was conducted only at major train stations.
- Commuters who do not use train stations were less likely to be included.

The student wants to emphasize a limitation of using the survey to represent all commuters in the city. Which choice most effectively uses relevant information from the notes?

- A** Of the 600 commuters surveyed, 48% preferred expanded bus service.
- B** Major train stations provide access to many commuters in the city.
- C** The survey asked commuters about expanded bus service rather than train service.
- D** Because the survey sampled commuters at major train stations, it may underrepresent commuters who do not use those stations.

Reading & Writing · Module 1 · Question 25

Worked solution

Answer D

NOTES SYNTHESIS AND LIMITATIONS

- 1 The survey sampled commuters at a train station.
- 2 People who do not use that station could be missing from the sample, so the result may underrepresent nonusers.

Reasoning check The limitation follows from the location-based sampling method.

Question 26

Reading & Writing · Module 1 · Diagnostic Test 6

Expression of Ideas

Multiple choice

A student has taken these notes:

- In 2019, engineers identified excessive vibration in a pedestrian bridge.
- In 2020, they installed temporary dampers.
- Measurements after installation showed lower vibration.
- In 2022, permanent dampers replaced the temporary system.

The student wants to present the sequence of the engineers' response. Which choice most effectively uses relevant information from the notes?

- A** After identifying excessive vibration in 2019, engineers installed temporary dampers in 2020 and replaced them with permanent dampers in 2022 after measurements showed reduced vibration.
- B** Engineers studied a pedestrian bridge in 2019, 2020, and 2022.
- C** Permanent dampers, installed in 2022, can reduce vibration in pedestrian bridges.
- D** Measurements showed that the temporary dampers installed by engineers reduced bridge vibration.

Reading & Writing · Module 1 · Question 26

Worked solution

Answer **A**

SEQUENCE AND CHRONOLOGY

- 1 The events must occur in the order given by the notes: the 2019 observation, the 2020 change, and the 2022 result.
- 2 The selected sentence includes all three dates in that chronological and causal sequence.

Reasoning check The best choice preserves both the timeline and the connection between the events.

Question 27

Reading & Writing · Module 1 · Diagnostic Test 6

Expression of Ideas

Multiple choice

A student has taken these notes about sculptor Lila Moreno:

- Moreno is known for large public sculptures made from reclaimed steel.
- Her 2018 work *Current* was built from steel recovered from a demolished factory.
- *Current* was installed beside the river in Moreno's hometown.
- Critics have interpreted the work as linking industrial history with environmental renewal.

The student is writing for readers already familiar with Moreno and wants to emphasize the significance of *Current*. Which choice most effectively uses relevant information from the notes?

- A** *Current*, a 2018 work by Moreno, is installed beside a river in her hometown.
- B** Critics have written about Moreno's work, including the sculpture *Current*.
- C** By using steel from a demolished factory in a riverside installation, *Current* connects the city's industrial past with ideas of environmental renewal.
- D** Lila Moreno is a sculptor known for creating large public works from reclaimed steel.

Reading & Writing · Module 1 · Question 27

Worked solution

Answer **C**

NOTES SYNTHESIS

- 1 The steel came from an industrial site, and the project placed it beside a river as part of an environmental renewal effort.
- 2 The sentence linking the industrial past with the riverside restoration uses both central details.

Reasoning check The correct synthesis includes the material's former use and its new ecological setting.



SAT DIAGNOSTIC TEST 6

Reading & Writing · Module 2

27 questions

32 min

No calculator

Questions for the Reading & Writing · Module 2 section. Craft and Structure, Information and Ideas, Standard English Conventions, and Expression of Ideas.

Question 1

Reading & Writing · Module 2 · Diagnostic Test 6

Craft and Structure

Multiple choice

Although the new dating technique produced a much narrower age range for the artifact, the researchers remained _____ in their interpretation because the surrounding sediment had been disturbed centuries earlier.

Which choice completes the text with the most logical and precise word or phrase?

A dismissive

B indifferent

C circumspect

D categorical

Reading & Writing · Module 2 · Question 1

Worked solution

Answer C

VOCABULARY IN CONTEXT

1 Because the sediment had been disturbed, the researcher avoids making a confident claim.

2 “Circumspect” means cautious and careful, which fits that attitude.

Reasoning check The context calls for caution, not certainty or speed.

Question 2

Reading & Writing · Module 2 · Diagnostic Test 6

Craft and Structure

Multiple choice

Two climate records from neighboring valleys appear to conflict: one indicates a wet century while the other indicates drought. Geologist Priya Nair argues that the discrepancy can be _____ if the first record is understood to reflect winter precipitation and the second summer precipitation.

Which choice completes the text with the most logical and precise word or phrase?

- ☐ A reconciled
- ☐ B amplified
- ☐ C prescribed
- ☐ D circumvented

Reading & Writing · Module 2 · Question 2

Worked solution

Answer **A**

VOCABULARY IN CONTEXT

- 1 The two records seem to disagree at first.
- 2 The seasonal distinction explains how both can be accurate, so the discrepancy can be reconciled.

Reasoning check To reconcile evidence is to make apparently conflicting evidence consistent.

Question 3

Reading & Writing · Module 2 · Diagnostic Test 6

Craft and Structure

Multiple choice

Some analysts claim that remote work inevitably reduces collaboration because employees encounter one another less often. That claim is plausible for teams whose coordination depends on frequent unplanned conversation. But studies of teams that use structured check-ins have found little difference in collaboration between remote and in-person groups.

Which choice best describes the function of the underlined sentence in the text as a whole?

- ☐ A It supplies evidence proving that remote teams collaborate less effectively.
- ☐ B It introduces the structured practices that the later sentence evaluates.
- ☐ C It restates the text's final conclusion in more cautious language.
- ☐ D It concedes that the opposing claim may apply under a specific condition before the text limits that claim.

Reading & Writing · Module 2 · Question 3

Worked solution

Answer **D**

AUTHOR'S PURPOSE

- 1 The sentence first acknowledges that the opposing claim could be true under a particular condition.
- 2 The following qualification limits how broadly that claim should be applied.

Reasoning check The structure concedes a point and then narrows its significance.

Question 4

Reading & Writing · Module 2 · Diagnostic Test 6

Craft and Structure

Multiple choice

Text 1

Digital restoration can reveal details in damaged paintings that viewers could not otherwise see. When restorers clearly distinguish reconstructed regions from surviving paint in documentation, digital additions can be valuable interpretive tools.

Text 2

Because viewers tend to treat polished digital reconstructions as authoritative, even carefully documented restorations can encourage unwarranted confidence about uncertain features. Reconstructions should therefore display uncertainty within the image itself.

How would the author of Text 2 most likely respond to Text 1's claim about documentation?

- ☐ A Digital restoration is valuable only when no part of the original painting survives.
- ☐ B Documentation eliminates the risk that viewers will mistake reconstructed details for surviving paint.
- ☐ C Documentation is useful but insufficient if the reconstruction itself does not communicate uncertainty to viewers.
- ☐ D Restorers should avoid recording how reconstructed regions differ from original paint.

Reading & Writing · Module 2 · Question 4

Worked solution

Answer C

PAIRED-TEXT COMPARISON

- 1 The authors agree that documenting a reconstruction can be useful.
- 2 The disagreement is about whether the image also needs to show the uncertainty surrounding the reconstruction; Text 2 says it does.

Reasoning check Documentation is helpful but insufficient when uncertainty is hidden.

Question 5

Reading & Writing · Module 2 · Diagnostic Test 6

Craft and Structure

Multiple choice

The dormant seeds remained viable for decades, revealing a _____ capacity for germination that became evident only after the soil was disturbed and exposed to light.

Which choice completes the text with the most logical and precise word or phrase?

A incessant

B negligible

C derivative

D latent

Reading & Writing · Module 2 · Question 5

Worked solution

Answer D

VOCABULARY IN CONTEXT

- 1 The passage describes a capacity that is present but not yet expressed.
- 2 “Latent” means hidden or not currently developed, so it fits the sentence.

Reasoning check The word refers to potential that has not yet appeared.

Question 6

Reading & Writing · Module 2 · Diagnostic Test 6

Craft and Structure

Multiple choice

Economist Jo Park compares a city's transportation network to a braided river: traffic can shift among several routes when one route is blocked. **The comparison is not meant to imply that travelers move randomly; rather, it highlights the network's ability to redistribute flow.** Park uses this idea to explain why closing one street may have smaller effects than a model based on a single route predicts.

Which choice best describes the function of the underlined sentence in the text as a whole?

- ☐ A It introduces evidence showing that travelers choose routes randomly.
- ☐ B It argues that route closures never substantially affect traffic.
- ☐ C It clarifies the intended point of an analogy while rejecting a possible misinterpretation of it.
- ☐ D It explains why braided rivers are more predictable than transportation networks.

Reading & Writing · Module 2 · Question 6

Worked solution

Answer C

FUNCTION OF AN ANALOGY

- 1 The river analogy makes the idea of water redistributing through a channel easier to visualize.
- 2 It clarifies an organized process and prevents readers from interpreting the movement as random.

Reasoning check The analogy explains the mechanism rather than merely adding a decorative comparison.

Question 7

Reading & Writing · Module 2 · Diagnostic Test 6

Craft and Structure

Multiple choice

Text 1

Chemical signatures in pottery can identify the geological region where its clay originated. When pottery found far from that region shares the signature, the finding provides strong evidence of long-distance exchange.

Text 2

Matching chemical signatures establish that material moved between regions, but not necessarily through trade. Migrants may have carried pottery with them, or finished objects may have changed hands repeatedly without organized exchange.

Based on the texts, the author of Text 2 would most likely agree with Text 1 that chemical signatures can establish which fact?

- ☐ A Organized trade was the only mechanism by which pottery moved.
- ☐ B Pottery or its material moved from one region to another.
- ☐ C The people who made the pottery also transported it to its final location.
- ☐ D Pottery moved directly between its source region and the site where it was found.

Reading & Writing · Module 2 · Question 7

Worked solution

Answer **B**

PAIRED-TEXT COMPARISON

- 1 Chemical signatures can match material in one region to material found in another.
- 2 That evidence establishes movement between regions, but it does not identify the exact mechanism or route.

Reasoning check The evidence supports where the material traveled, not why or how it traveled.

Question 8

Reading & Writing · Module 2 · Diagnostic Test 6

Information and Ideas

Multiple choice

On islands where a predatory lizard was introduced, one insect species became less abundant near the ground but more abundant in tree canopies. Individuals collected from canopy populations also had larger adhesive pads than individuals from islands without the lizard. Because the shift occurred over several generations, researchers caution that behavior and natural selection may both have contributed.

Which inference is best supported by the text?

- ☐ A Canopy insects on islands without the lizard have larger adhesive pads than ground insects.
- ☒ B The predator's presence is associated with both a change in where the insects are found and a difference in a trait that could aid canopy living.
- ☐ C The insects moved into tree canopies entirely because individuals learned to avoid the ground.
- ☐ D Larger adhesive pads caused the predatory lizard to be introduced to the islands.

Reading & Writing · Module 2 · Question 8

Worked solution

Answer B

INFERENCE FROM RESEARCH FINDINGS

- 1 The predator is associated with the location shift, and the shifted population has larger adhesive pads.
- 2 Those observations support a connection between the predator's presence, location, and pad size, while not by themselves proving natural selection.

Reasoning check The answer must state the observed association without turning it into an unsupported causal claim.

Question 9

Reading & Writing · Module 2 · Diagnostic Test 6

Information and Ideas

Multiple choice

Some old-growth forests store large amounts of carbon not only because their trees are large but also because decomposition can be unusually slow in cool, wet soils. A warming climate may therefore affect stored carbon through two pathways: by changing tree growth and by accelerating the release of carbon from soil organic matter.

Which choice best states the main idea of the text?

- ☐ A Warming is likely to increase tree growth enough to offset all carbon released from soils.
- ☐ B Climate warming can alter forest carbon storage by affecting both carbon accumulation in trees and carbon release from soils.
- ☐ C Old-growth forests store carbon primarily because decomposition stops completely in wet soils.
- ☐ D Soil organic matter contributes little to carbon storage in old-growth forests.

Reading & Writing · Module 2 · Question 9

Worked solution

Answer **B**

MAIN IDEA

- 1 The passage describes warming as affecting both how trees accumulate and how carbon is released from soil.
- 2 Together, those pathways explain why the forest's carbon balance may change.

Reasoning check The main idea includes both the vegetation and soil-carbon effects.

Question 10

Reading & Writing · Module 2 · Diagnostic Test 6

Information and Ideas

Multiple choice

A laboratory measured the output of three solar-cell designs before and after a high-temperature stress test.

Design	Before (W)	After (W)
A	100	92
B	80	76
C	120	105

A researcher claims that Design B was the most resistant to the stress test when resistance is measured by percentage of original output retained.

Which choice best uses the data to support the researcher's claim?

- ☐ A B retained 95% of its original output, compared with 92% for A and 87.5% for C.
- ☐ B C lost 15 W, while B lost only 4 W.
- ☐ C A began at 100 W and ended at 92 W.
- ☐ D B produced 76 W after the test, less than either A or C.

Reading & Writing · Module 2 · Question 10

Worked solution

Answer A

DATA INTERPRETATION

- 1 For A, the retained percentage is $92 \div 100 = 92\%$.
- 2 For B it is $76 \div 80 = 95\%$, and for C it is $105 \div 120 = 87.5\%$.

Reasoning check B has the greatest retained percentage.

Question 11

Reading & Writing · Module 2 · Diagnostic Test 6

Information and Ideas

Multiple choice

A cognitive scientist proposes that short retrieval quizzes improve later memory because retrieving information strengthens access to it, not simply because quizzes expose students to the material again.

Which experiment would best test the proposed mechanism?

- A** Give all students the same retrieval quiz and compare their scores on that quiz.
- B** Give one group a retrieval quiz and ask whether the students enjoyed it.
- C** Compare students who retrieve answers from memory with students who reread the same questions and answers for the same amount of time, then give both groups a delayed test.
- D** Compare students who take a retrieval quiz with students who study an unrelated subject for less time.

Reading & Writing · Module 2 · Question 11

Worked solution

Answer C

EXPERIMENTAL DESIGN

- 1** The retrieval group spends the same amount of time as the rereading group, so time is controlled.
- 2** Testing everyone after a delay then compares later performance while isolating the effect of retrieval practice.

Reasoning check The design is meant to test retrieval versus rereading, not unequal study time.

Question 12

Reading & Writing · Module 2 · Diagnostic Test 6

Information and Ideas

Multiple choice

A researcher argues that a new irrigation system increased crop yield because farms adopting the system produced more grain than farms that did not. However, the adopting farms were also, on average, located in regions with substantially richer soil.

Which finding would most weaken the researcher's causal claim?

- ☐ A The new irrigation system requires less water than the older system.
- ☐ B Farmers reported that the new system was easier to maintain.
- ☐ C Among farms with similar soil quality, adopters and nonadopters had nearly identical yields.
- ☐ D Several adopting farms produced more grain than they had the previous year.

Reading & Writing · Module 2 · Question 12

Worked solution

Answer C

CAUSAL INFERENCE

- 1 Matching adopters and nonadopters on soil type removes an important possible confounding factor.
- 2 If yields remain similar after that match, the evidence weakens the claim that irrigation alone caused higher yields.

Reasoning check The comparison makes the proposed cause less persuasive because the outcome does not differ.

Question 13

Reading & Writing · Module 2 · Diagnostic Test 6

Information and Ideas

Multiple choice

When the committee finally approved Mara's proposal, she thanked each member with careful formality. Outside the room, her colleague expected celebration, but Mara merely checked the time and said, "Good. Now we can find out whether it works."

Which choice best describes Mara's attitude toward the committee's approval?

- ☐ A She values the approval mainly because it allows the proposal to be tested.
- ☐ B She doubts that the committee members understood the proposal.
- ☐ C She considers approval more important than the proposal's eventual results.
- ☐ D She is disappointed that the committee approved the proposal too quickly.

Reading & Writing · Module 2 · Question 13

Worked solution

Answer **A**

AUTHOR'S ATTITUDE

- 1 The author treats the approval as valuable because it creates an opportunity to test the proposed design.
- 2 The tone is interested and cautiously positive, not a claim that approval proves success.

Reasoning check Approval matters as access to a test, not as evidence that the hypothesis is already confirmed.

Question 14

Reading & Writing · Module 2 · Diagnostic Test 6

Information and Ideas

Multiple choice

Two groups of 40 seedlings were grown under different light conditions. Group X had a median height of 18 cm and an interquartile range of 4 cm. Group Y had a median height of 18 cm and an interquartile range of 9 cm.

Which statement is best supported by the information?

- ☐ A Group Y has a greater mean height than Group X.
- ☐ B Every seedling in Group X is closer to 18 cm than every seedling in Group Y.
- ☐ C Group X and Group Y have identical height distributions because their medians are equal.
- ☐ D The middle half of Group Y's heights is more spread out than the middle half of Group X's heights.

Reading & Writing · Module 2 · Question 14

Worked solution

Answer **D**

DATA INTERPRETATION

- 1 The interquartile range is the span of the middle half of the observations.
- 2 The given IQRs are 4 cm for X and 9 cm for Y, so the middle half of Y's heights is more spread out.

Reasoning check Y has the larger interquartile range.

Question 15

Reading & Writing · Module 2 · Diagnostic Test 6

Standard English Conventions

Multiple choice

The researchers initially compared the variables without separating participants by age. The first analysis suggested that the two variables were strongly related _____ however, when the researchers controlled for age, the relationship became much weaker.

Which choice completes the text so that it conforms to the conventions of Standard English?

A : however,

B ; however,

C however;

D , however,

Reading & Writing · Module 2 · Question 15

Worked solution

Answer **B**

PUNCTUATION

- 1 The words on each side of the blank are independent clauses.
- 2 Join them with a semicolon, then set off the conjunctive adverb “however” with a comma.

Reasoning check The correct pattern is “; however,”.

Question 16

Reading & Writing · Module 2 · Diagnostic Test 6

Standard English Conventions

Multiple choice

The museum's autumn exhibition will feature several works that combine traditional materials with digital media. Rina Patel's newest installation, _____ will open at the museum in October.

Which choice completes the text so that it conforms to the conventions of Standard English?

- ☐ A which combines projected images with handwoven fabric
- ☐ B that combines projected images with handwoven fabric
- ☐ C which combines projected images with handwoven fabric,
- ☐ D that combines projected images with handwoven fabric,

Reading & Writing · Module 2 · Question 16

Worked solution

Answer C

RELATIVE CLAUSES

- 1 The clause beginning with “which” gives extra information about a specific installation.
- 2 Because it is nonessential, it needs a comma before and after the clause.

Reasoning check The nonessential relative clause is set off with paired commas.

Question 17

Reading & Writing · Module 2 · Diagnostic Test 6

Standard English Conventions

Multiple choice

The original classification system placed several visually similar species in the same category despite differences in behavior and habitat. After reviewing more than 2,000 observations, _____.

Which choice completes the text so that it conforms to the conventions of Standard English?

- ☐ A the classification system was revised by the research team
- ☐ B there was a revision to the research team's classification system
- ☐ C a revision of the classification system was made
- ☐ D the research team revised its classification system

Reading & Writing · Module 2 · Question 17

Worked solution

Answer D

MODIFIER PLACEMENT

- 1 “After reviewing the records” must describe the people who reviewed them.
- 2 The research team is the logical subject, so placing it immediately after the modifier avoids a dangling modifier.

Reasoning check The team, not the records, performs the reviewing.

Question 18

Reading & Writing · Module 2 · Diagnostic Test 6

Standard English Conventions

Multiple choice

Most official reports describe only the expedition's final route. The collection of letters, field notes, and unpublished sketches _____ historians an unusually detailed view of the expedition's planning.

Which choice completes the text so that it conforms to the conventions of Standard English?

A have given

B are giving

C give

D gives

Reading & Writing · Module 2 · Question 18

Worked solution

Answer D

SUBJECT-VERB AGREEMENT

1 The subject is “the collection,” which is singular.

2 The correct verb is “gives”; the plural noun “letters” does not control agreement.

Reasoning check The main subject determines the singular verb.

Question 19

Reading & Writing · Module 2 · Diagnostic Test 6

Standard English Conventions

Multiple choice

A surprising effect first appeared in a small pilot study. The two laboratories independently reproduced the _____ results, strengthening confidence that the effect was not unique to one research group.

Which choice completes the text so that it conforms to the conventions of Standard English?

A original studies

B original study

C original studies'

D original study's

Reading & Writing · Module 2 · Question 19

Worked solution

Answer D

POSSESSIVE NOUNS

- 1 The results belong to one original study.
- 2 The singular possessive form is “the original study’s results.”

Reasoning check The apostrophe and s mark possession by one study.

Question 20

Reading & Writing · Module 2 · Diagnostic Test 6

Standard English Conventions

Multiple choice

Researchers can trace how the design changed from proposal to construction. The archive contains evidence from three stages of the project _____ early sketches, revised plans, and photographs of the completed structure.

Which choice completes the text so that it conforms to the conventions of Standard English?

A ;

B :

C , and

D . Although

Reading & Writing · Module 2 · Question 20

Worked solution

Answer **B**

PUNCTUATION AND LISTS

1 The independent clause before the blank introduces examples that follow.

2 A colon is the punctuation mark that correctly introduces the list.

Reasoning check A colon follows the complete introductory clause.

Question 21

Reading & Writing · Module 2 · Diagnostic Test 6

Standard English Conventions

Multiple choice

The expeditions were scheduled several weeks apart so that equipment could be established in advance. Before the second expedition reached the site, the first team _____ a weather station there and had begun transmitting daily measurements.

Which choice completes the text so that it conforms to the conventions of Standard English?

A installs

B had installed

C will install

D is installing

Reading & Writing · Module 2 · Question 21

Worked solution

Answer **B**

SEQUENCE OF TENSES

1 The equipment was installed before the second expedition arrived.

2 The past perfect “had installed” identifies the earlier past action.

Reasoning check The earlier action takes the past-perfect form.

Question 22

Reading & Writing · Module 2 · Diagnostic Test 6

Expression of Ideas

Multiple choice

Engineers are evaluating two alloys for equipment that must operate in polar conditions. Both alloys have nearly the same density. Alloy P remains flexible at low temperatures; _____ Alloy Q becomes brittle below -20°C .

Which choice completes the text with the most logical transition?

A for example,

B by contrast,

C therefore,

D likewise,

Reading & Writing · Module 2 · Question 22

Worked solution

Answer B

TRANSITIONS

1 Alloy P remains flexible at low temperatures, whereas Alloy Q becomes brittle below -20°C .

2 “By contrast” correctly marks the difference between the two alloys.

Reasoning check The transition should mark a contrast between the materials.

Question 23

Reading & Writing · Module 2 · Diagnostic Test 6

Expression of Ideas

Multiple choice

The original algorithm repeats several operations whenever a new data point is added. The revised algorithm requires fewer calculations for each data point. _____ it can process the same data set in substantially less time.

Which choice completes the text with the most logical transition?

A Consequently,

B Specifically,

C Meanwhile,

D Nevertheless,

Reading & Writing · Module 2 · Question 23

Worked solution

Answer **A**

TRANSITIONS

1 The revised algorithm requires fewer calculations for each data point.

2 The reduced calculation load means the algorithm can process the same data set in substantially less time, so “Consequently” correctly introduces the result.

Reasoning check The second statement follows as an effect of the first.

Question 24

Reading & Writing · Module 2 · Diagnostic Test 6

Expression of Ideas

Multiple choice

A student takes these notes:

- Some desert plants open leaf pores at night rather than during the day.
- Night air is generally cooler and more humid.
- Opening pores allows carbon dioxide to enter but also permits water to escape.
- Opening pores at night can reduce water loss.

The student wants to explain the mechanism by which nighttime pore opening conserves water. Which choice most effectively uses relevant information from the notes?

- A** Leaf pores allow carbon dioxide to enter a plant and can also permit water to escape.
- B** Desert plants often live where daytime air is hotter than nighttime air.
- C** By opening their pores when the air is cooler and more humid, some desert plants can take in carbon dioxide while losing less water.
- D** Some desert plants differ from other plants because their leaf pores open at night.

Reading & Writing · Module 2 · Question 24

Worked solution

Answer **C**

NOTES SYNTHESIS

- 1 The notes indicate that nighttime is cooler and more humid.
- 2 Taking in air at that time can reduce water loss while still supplying the needed intake.

Reasoning check The best synthesis includes both the timing and the water-conservation mechanism.

Question 25

Reading & Writing · Module 2 · Diagnostic Test 6

Expression of Ideas

Multiple choice

A student takes these notes:

- Researcher A measured river pollution using chemical samples collected once each month.
- Researcher B used sensors that recorded conductivity every fifteen minutes.
- Chemical samples identify specific pollutants.
- Conductivity sensors can reveal rapid changes but do not identify every pollutant.

The student wants to compare a strength of each approach. Which choice most effectively uses relevant information from the notes?

- A** Chemical samples and conductivity sensors are both methods for studying river pollution.
- B** Researcher A collected samples monthly, while Researcher B collected conductivity data every fifteen minutes.
- C** Researcher B's sensors cannot identify every pollutant that chemical samples can detect.
- D** Researcher A's sampling can identify specific pollutants, whereas Researcher B's frequent sensor readings can capture rapid changes.

Reading & Writing · Module 2 · Question 25

Worked solution

Answer **D**

NOTES SYNTHESIS

- 1 Samples identify which pollutants are present, while sensors can record rapid changes as conditions shift.
- 2 The two methods therefore provide complementary information rather than duplicating the same measurement.

Reasoning check The choice must preserve the distinct strength of each method.

Question 26

Reading & Writing · Module 2 · Diagnostic Test 6

Expression of Ideas

Multiple choice

A student has taken these notes about the telescope Helios:

- Helios began observations in 2024.
- Its mirror is segmented.
- It can combine data from its mirror segments to act as one larger reflecting surface.
- This design allows Helios to collect faint light from distant objects.

The student is writing for an audience already familiar with Helios and wants to emphasize why its segmented mirror matters. Which choice most effectively uses relevant information from the notes?

- A** Distant objects can emit light that is faint by the time it reaches a telescope.
- B** The mirror on Helios is made of multiple segments rather than a single piece.
- C** Helios, which began observations in 2024, is a telescope with a segmented mirror.
- D** By combining data from its mirror segments as though they formed one larger surface, Helios can collect faint light from distant objects.

Reading & Writing · Module 2 · Question 26

Worked solution

Answer **D**

NOTES SYNTHESIS

- 1 A segmented mirror uses several smaller pieces that work together.
- 2 Combining those segments creates the light-gathering surface of a much larger mirror, allowing the telescope to detect faint objects.

Reasoning check The key benefit is a larger effective collecting surface.

Question 27

Reading & Writing · Module 2 · Diagnostic Test 6

Expression of Ideas

Multiple choice

A student takes these notes:

- A manuscript was traditionally dated to the 1400s based on handwriting style.
- A recent analysis dated its parchment to 1320–1390.
- Parchment can be stored for years before being written on.
- The ink has not yet been independently dated.

The student wants to emphasize what remains uncertain about the manuscript's date of writing. Which choice most effectively uses relevant information from the notes?

- A** Although the parchment dates to 1320–1390, the manuscript could have been written later because parchment may be stored before use and the ink has not yet been dated.
- B** The parchment analysis provides a date range of 1320–1390.
- C** A recent analysis dated the parchment to a period earlier than the traditional date.
- D** The manuscript was traditionally assigned to the 1400s because of its handwriting style.

Reading & Writing · Module 2 · Question 27

Worked solution

Answer **A**

NOTES SYNTHESIS AND UNCERTAINTY

- 1 The parchment may have been made before the writing was added, and the ink itself has not been dated.
- 2 Those facts explain why the artifact's age does not necessarily establish the age of the writing.

Reasoning check The uncertainty comes from the gap between the parchment's date and the ink's unknown date.



SAT DIAGNOSTIC TEST 6

Math · Module 1

22 questions

35 min

Calculator allowed

Questions for the Math · Module 1 section. Algebra, Advanced Math, Problem-Solving and Data Analysis, and Geometry and Trigonometry.

Question 1

Math · Module 1 · Diagnostic Test 6

Algebra

Multiple choice

The system

$$2x + 3y = 13$$

$$5x - 2y = 4$$

If the solution is (x, y) , what is the value of $x + y$?

A 6

B 4

C 5

D 7

Math · Module 1 · Question 1

Worked solution

Answer C

LINEAR SYSTEMS

1 Multiply $2x + 3y = 13$ by 2 and $5x - 2y = 4$ by 3. Adding eliminates y : $19x = 38$, so $x = 2$.

2 Substitution gives $2(2) + 3y = 13$, hence $y = 3$. Therefore $x + y = 5$.

Reasoning check The ordered pair is $(2, 3)$, so the requested sum is 5.

Question 2

Math · Module 1 · Diagnostic Test 6

Problem-Solving and Data Analysis

Multiple choice

A fund invests 35% of its money in Asset A and the remaining 65% in Asset B. Over one year, Asset A earns a 12% return, and the fund as a whole earns a 7.45% return.

What is the annual return on Asset B?

A 5%

B 4%

C 6.5%

D 6%

Math · Module 1 · Question 2

Worked solution

Answer **A**

WEIGHTED RETURNS

1 The portfolio equation is $0.35(0.12) + 0.65r = 0.0745$, where r is Asset B's return.

2 Since $0.042 + 0.65r = 0.0745$, $0.65r = 0.0325$ and $r = 0.05$.

Reasoning check Asset B's annual return is 5%.

Question 3

Math · Module 1 · Diagnostic Test 6

Advanced Math

Multiple choice

An exponential function $f(x) = ab^x$, where $a > 0$ and $b > 1$, satisfies $f(1) = 18$ and $f(4) = 486$.

By what percent does $f(x)$ increase whenever x increases by 2?

A 800%

B 200%

C 900%

D 300%

Math · Module 1 · Question 3

Worked solution

Answer **A**

EXPONENTIAL GROWTH

1 For $f(x) = ab^x$, the ratio $f(4) \div f(1) = 486 \div 18 = 27$, so $b^3 = 27$ and $b = 3$.

2 Increasing x by 2 multiplies the value by $b^2 = 9$. The percent increase is $(9 - 1) \times 100\% = 800\%$.

Reasoning check A factor of 9 is an 800% increase, not a 9% increase.

Question 4

Math · Module 1 · Diagnostic Test 6

Algebra

Multiple choice

A linear function g satisfies $g(-2) = 5$ and $g(4) = -7$.

What is the x -intercept of the graph of $y = g(x)$?

A 1

B $\frac{1}{2}$

C $-\frac{1}{2}$

D 2

Math · Module 1 · Question 4

Worked solution

Answer B

LINEAR FUNCTIONS AND INTERCEPTS

1 The slope through $(-2, 5)$ and $(4, -7)$ is $(-7 - 5) \div (4 - (-2)) = -2$.

2 Using $5 = -2(-2) + b$ gives $b = 1$, so $g(x) = -2x + 1$. Setting $g(x) = 0$ gives $x = \frac{1}{2}$.

Reasoning check The graph crosses the x -axis at $(\frac{1}{2}, 0)$.

Question 5

Math · Module 1 · Diagnostic Test 6

Geometry and Trigonometry

Multiple choice

In a right triangle, the altitude from the right angle to the hypotenuse divides the hypotenuse into segments of lengths 9 and 16. Let θ be the smaller acute angle of the triangle.

What is $\tan \theta$?

A $\frac{4}{3}$

B $\frac{3}{4}$

C $\frac{4}{5}$

D $\frac{3}{5}$

Math · Module 1 · Question 5

Worked solution

Answer B

RIGHT-TRIANGLE SIMILARITY

- 1 The hypotenuse is $9 + 16 = 25$. By the altitude theorem, the legs have lengths $\sqrt{(25 \times 9)} = 15$ and $\sqrt{(25 \times 16)} = 20$.
- 2 For the smaller acute angle, use the shorter leg as the opposite side: $\tan \theta = 15 \div 20 = \frac{3}{4}$.

Reasoning check The tangent ratio reduces to $\frac{3}{4}$.

Question 6

Math · Module 1 · Diagnostic Test 6

Algebra

Multiple choice

The system

$$(k + 1)x - 3y = 7$$

$$4x - 6y = 5$$

has no solution.

What is the value of k ?

A -1

B 1

C 2

D 0

Math · Module 1 · Question 6

Worked solution

Answer B

PARALLEL LINES AND NO SOLUTION

- 1 The equations are $(k + 1)x - 3y = 7$ and $4x - 6y = 5$. For no solution, the coefficient pairs must be proportional while the constants are not.
- 2 Matching the y-coefficient requires multiplying the first equation by 2: $2(k + 1) = 4$. Thus $k = 1$, and the constants 14 and 5 are different.

Reasoning check At $k = 1$, the lines are parallel and distinct, so the system has no solution.

Question 7

Math · Module 1 · Diagnostic Test 6

Advanced Math

Multiple choice

The quadratic function q is defined by $q(x) = a(x - 4)^2 - 18$. If $q(1) = 0$ and r and s are the x -intercepts of the graph of $y = q(x)$,

what is the value of $r^2 + s^2$?

A 42

B 46

C 50

D 58

Math · Module 1 · Question 7

Worked solution

Answer C

QUADRATIC VERTEX FORM

1 Given $q(x) = a(x - 4)^2 - 18$ and $q(1) = 0$, $9a - 18 = 0$, so $a = 2$.

2 The roots satisfy $2(x - 4)^2 - 18 = 0$, so $x = 1$ or $x = 7$. Hence $r^2 + s^2 = 1^2 + 7^2 = 50$.

Reasoning check The two roots are 1 and 7, giving a square-sum of 50.

Question 8

Math • Module 1 • Diagnostic Test 6

Problem-Solving and Data Analysis

Multiple choice

A study records whether 60 students practiced more than 5 hours and whether they scored at least 14 points.

	14 or more	Below 14
More than 5 hours	24	6
5 hours or less	10	20

A student is selected at random from those who scored at least 14 points. What is the probability that the student practiced more than 5 hours?

A $\frac{3}{4}$

B $\frac{12}{17}$

C $\frac{4}{5}$

D $\frac{2}{5}$

Math • Module 1 • Question 8

Worked solution

Answer **B**

CONDITIONAL PROBABILITY

- 1 Among students scoring at least 14, the total is $10 + 24 = 34$.
- 2 Of those students, 24 practiced more than 5 hours, so the probability is $24 \div 34 = \frac{12}{17}$.

Reasoning check The denominator is the restricted group of 34 students, not the full table.

Question 9

Math • Module 1 • Diagnostic Test 6

Algebra

Multiple choice

A rideshare company charges a fixed booking fee plus \$1.80 per mile for the first 10 miles and \$1.20 per mile for each mile beyond 10. A 6-mile trip costs \$15.30.

How much does an 18-mile trip cost under the same pricing rule?

A \$30.90

B \$33.30

C \$31.50

D \$32.10

Math • Module 1 • Question 9

Worked solution

Answer D

PIECEWISE PRICING

1 A 6-mile trip costs $F + 1.80(6) = 15.30$, so the fixed fee is $F = 4.50$.

2 An 18-mile trip includes the first 10 miles at 1.80 per mile and the remaining 8 at 1.20: $4.50 + 1.80(10) + 1.20(8) = 32.10$.

Reasoning check The total charge is \$32.10.

Question 10

Math • Module 1 • Diagnostic Test 6

Advanced Math

Multiple choice

For $x \neq -3$ and $x \neq 3$, consider

$$\frac{\frac{x}{x-3} + \frac{3}{3-x}}{\frac{x^2-9}{x+3}}.$$

Which expression is equivalent to the given expression?

A $\frac{1}{x-3}$

B $\frac{x}{x^2-9}$

C $\frac{1}{x+3}$

D $x - 3$

Math • Module 1 • Question 10

Worked solution

Answer A

RATIONAL EXPRESSIONS

1 Because $3 - x = -(x - 3)$, the numerator becomes $x \div (x - 3) - 3 \div (x - 3) = 1$.

2 The denominator is $(x^2 - 9) \div (x + 3) = x - 3$, so the whole expression is $1 \div (x - 3)$.

Reasoning check The restrictions $x \neq -3$ and $x \neq 3$ remain in force after simplification.

Question 11

Math • Module 1 • Diagnostic Test 6

Geometry and Trigonometry

Multiple choice

The equation

$$x^2 + y^2 - 6x + 8y - 11 = 0$$

represents a circle. A square is circumscribed about the circle so that each side of the square is tangent to the circle.

What is the area of the square?

A 36

B 144

C 100

D 72

Math • Module 1 • Question 11

Worked solution

Answer B

COMPLETING THE SQUARE

- 1 Rewrite $x^2 - 6x + y^2 + 8y - 11 = 0$ as $(x - 3)^2 + (y + 4)^2 = 36$. The circle's radius is 6.
- 2 A square circumscribed about the circle has side length equal to the diameter, 12. Its area is $12^2 = 144$.

Reasoning check Because the square surrounds the circle, its side is the circle's diameter, not the circle's diagonal.

Question 12

Math · Module 1 · Diagnostic Test 6

Algebra

Multiple choice

The line

$$2x - 5y = 10$$

is shifted 3 units to the right and 4 units down.

What is the x -coordinate of the x -intercept of the shifted line?

A 21

B 15

C 18

D 12

Math · Module 1 · Question 12

Worked solution

Answer C

TRANSLATIONS OF A LINE

1 For $2x - 5y = 10$, shifting right 3 replaces x by $x - 3$, and shifting down 4 replaces y by $y + 4$.

2 Thus $2(x - 3) - 5(y + 4) = 10$, which simplifies to $2x - 5y = 36$. Setting $y = 0$ gives $x = 18$.

Reasoning check The shifted line crosses the x -axis at $x = 18$.

Question 13

Math · Module 1 · Diagnostic Test 6

Advanced Math

Multiple choice

The polynomial $p(x) = 2x^3 + ax^2 + bx - 12$, where a and b are constants, is divisible by both $x - 2$ and $x + 1$.

What is the value of $a - b$?

A 6

B 14

C 18

D 10

Math · Module 1 · Question 13

Worked solution

Answer B

FACTOR THEOREM

1 Since $x - 2$ is a factor, $p(2) = 0$: $16 + 4a + 2b - 12 = 0$, so $2a + b = -2$.

2 Since $x + 1$ is a factor, $p(-1) = 0$: $-2 + a - b - 12 = 0$, so $a - b = 14$.

Reasoning check The second factor condition directly gives the requested value $a - b = 14$.

Question 14

Math • Module 1 • Diagnostic Test 6

Problem-Solving and Data Analysis

Multiple choice

A random sample estimates that 62% of a population has a certain characteristic, with a margin of error of 4 percentage points.

Which statement is best supported by this information?

- ☐ A The true population percentage must be between 58% and 66%.
- ☐ B Exactly 4% of the sampled individuals were measured incorrectly.
- ☐ C A population value of 60% is more consistent with the sample result than a population value of 69%.
- ☐ D Every other random sample of the same size must produce an estimate between 58% and 66%.

Math • Module 1 • Question 14

Worked solution

Answer C

MARGIN OF ERROR

- 1 The estimate is 62% with a margin of error of 4 percentage points, so the reported interval is 58% to 66%.
- 2 A population value of 60% lies in that interval, whereas 69% does not. The margin does not make the interval certain for every sample.

Reasoning check The information makes 60% more consistent with the estimate than 69%.

Question 15

Math · Module 1 · Diagnostic Test 6

Algebra

Multiple choice

A delivery truck carries two types of crates. Type A crates weigh 12 kilograms each, and Type B crates weigh 8 kilograms each. The truck must carry at least 60 crates, including at least 20 Type B crates, and the total crate weight can be at most 680 kilograms.

What is the greatest possible number of Type A crates the truck can carry?

A 40

B 43

C 42

D 41

Math · Module 1 · Question 15

Worked solution

Answer **B**

LINEAR CONSTRAINTS

- 1 Let x be Type A crates and y be Type B crates. The constraints include $x + y \geq 60$, $y \geq 20$, and $12x + 8y \leq 680$.
- 2 To maximize x , use the smallest possible y , namely 20: $12x + 160 \leq 680$, so $x \leq 43\frac{1}{3}$. The greatest whole number is 43.

Reasoning check The truck can carry at most 43 Type A crates.

Question 16

Math · Module 1 · Diagnostic Test 6

Advanced Math

Multiple choice

The equation

$$x^2 + (m - 6)x + 2m = 0$$

has two real solutions whose difference is 4.

What is the sum of all possible values of m ?

A 12

B 16

C 20

D 24

Math · Module 1 · Question 16

Worked solution

Answer C

DISCRIMINANT AND ROOT DIFFERENCE

1 For $x^2 + (m - 6)x + 2m = 0$, the roots differ by 4. If the roots are r and s , then $(r - s)^2 = (r + s)^2 - 4rs = 16$.

2 Vieta's formulas give $(m - 6)^2 - 8m = 16$, or $m^2 - 20m + 20 = 0$. The sum of the possible m -values is 20.

Reasoning check The sum of roots of the equation in m is 20.

Question 17

Math · Module 1 · Diagnostic Test 6

Advanced Math

Multiple choice

An exponential function $h(x) = ab^x$, where $a > 0$ and $b > 0$, satisfies $h(1) = 20$ and $h(4) = 540$.

What is the value of $h(-1)$?

A $\frac{20}{3}$

B 20

C $\frac{20}{9}$

D $\frac{20}{27}$

Math · Module 1 · Question 17

Worked solution

Answer C

EXPONENTIAL FUNCTIONS

1 For $h(x) = ab^x$, $h(4) \div h(1) = 540 \div 20 = 27$, so $b^3 = 27$ and $b = 3$.

2 Then $a = 20 \div 3 = \frac{20}{3}$, and $h(-1) = a \div 3 = \frac{20}{9}$.

Reasoning check The function value is $\frac{20}{9}$.

Question 18

Math · Module 1 · Diagnostic Test 6

Algebra

SPR

The equations

$$2(x + y) + 3(x - y) = 13$$

$$4(x + y) - (x - y) = 19$$

have solution (x, y) .

What is the value of $4xy$?



Student-produced response · enter a numerical answer

Math · Module 1 · Question 18

Worked solution

Answer **24**

LINEAR SYSTEMS WITH SUBSTITUTION VARIABLES

1

Let $u = x + y$ and $v = x - y$. The system becomes $2u + 3v = 13$ and $4u - v = 19$.

2

From the second equation, $v = 4u - 19$. Substitution gives $u = 5$ and $v = 1$. Thus $x = 3$, $y = 2$, and $4xy = 24$.

Reasoning check The transformed variables recover $x = 3$ and $y = 2$, so the requested value is 24.

Question 19

Math · Module 1 · Diagnostic Test 6

Advanced Math

SPR

The quadratic equation

$$x^2 - 10x + c = 0$$

has roots r and s . If $r^2 + s^2 = 58$,

what is the value of c ?



Student-produced response · enter a numerical answer

Math • Module 1 • Question 19

Worked solution

Answer 21

VIETA'S FORMULAS

- 1 For $x^2 - 10x + c = 0$ with roots r and s , Vieta gives $r + s = 10$ and $rs = c$.
- 2 Since $r^2 + s^2 = (r + s)^2 - 2rs = 58$, $100 - 2c = 58$, so $c = 21$.

Reasoning check The constant term is $c = 21$.

Question 20

Math • Module 1 • Diagnostic Test 6

Geometry and Trigonometry

SPR

A sphere has surface area 100π . The sphere is inscribed in a cube, and a second sphere passes through all eight vertices of the cube.

What is the square of the radius of the second sphere?

 **Student-produced response • enter a numerical answer**

Math • Module 1 • Question 20

Worked solution

Answer 75

SPHERE AND CUBE GEOMETRY

- 1 The sphere's surface area is $4\pi r^2 = 100\pi$, so $r^2 = 25$ and $r = 5$. The inscribed cube therefore has side length 10.
- 2 The second sphere passes through the cube's vertices, so its radius is half the cube's space diagonal: $(10\sqrt{3}) \div 2 = 5\sqrt{3}$. Its radius squared is 75.

Reasoning check The square of the second sphere's radius is 75.

Question 21

Math · Module 1 · Diagnostic Test 6

Algebra

SPR

The function f is linear. It satisfies $f(2) = 3f(-1)$ and $f(5) = 27$.

What is the value of $f(0)$?



Student-produced response · enter a numerical answer

Math · Module 1 · Question 21

Worked solution

Answer 9

LINEAR FUNCTIONS

1

Write $f(x) = mx + b$. The relation $f(2) = 3f(-1)$ gives $2m + b = 3(-m + b)$, so $b = 5m \div 2$.

2

Using $f(5) = 27$ gives $5m + 5m \div 2 = 27$, so $m = 18/5$ and $b = 9$. Therefore $f(0) = 9$.

Reasoning check The y-intercept of the linear function is 9.

Question 22

Math · Module 1 · Diagnostic Test 6

Advanced Math

SPR

The equation

$$|7 - 4x| = 2x + 5$$

has two real solutions.

What is the product of the two solutions?



Student-produced response · enter a numerical answer

Math • Module 1 • Question 22

Worked solution

Answer 2

ABSOLUTE-VALUE EQUATIONS

- 1 Solve $7 - 4x = 2x + 5$, which gives $x = \frac{1}{3}$.
- 2 For the negative case, $7 - 4x = -(2x + 5)$, giving $x = 6$. Their product is $\frac{1}{3} \times 6 = 2$.

Reasoning check The two solutions are $\frac{1}{3}$ and 6, so their product is 2.



SAT DIAGNOSTIC TEST 6

Math · Module 2

22 questions

35 min

Calculator allowed

Questions for the Math · Module 2 section. Algebra, Advanced Math, Problem-Solving and Data Analysis, and Geometry and Trigonometry.

Question 1

Math · Module 2 · Diagnostic Test 6

Algebra

Multiple choice

The system

$$2x + 5y = 16$$

$$(k - 1)x + (2k + 1)y = 4k + 8$$

has a unique solution (x, y) . If $x - y = 1$,

what is the value of k ?

A 3

B 1

C 5

D -3

Math · Module 2 · Question 1

Worked solution

Answer **A**

LINEAR SYSTEMS

1

From $x - y = 1$, write $x = y + 1$. Substitution into $2x + 5y = 16$ gives $y = 2$ and $x = 3$.

2

Substitute into the second equation: $3(k - 1) + 2(2k + 1) = 4k + 8$, so $7k - 1 = 4k + 8$ and $k = 3$.

Reasoning check The determinant is $7 - k = 4$, so the stated solution is unique and $k = 3$.

Question 2

Math · Module 2 · Diagnostic Test 6

Advanced Math

Multiple choice

The polynomial $r(x) = (x - 2)(x + 4)(x - 7)$. A function s is defined by $s(x) = r(3x + 1)$. If a, b, c are the three zeros of s ,

what is the value of abc ?

A $-\frac{10}{3}$

B $-\frac{10}{9}$

C $\frac{10}{3}$

D $\frac{10}{9}$

Math · Module 2 · Question 2

Worked solution

Answer B

ZEROS UNDER A TRANSFORMATION

- 1 The zeros of $r(x)$ are 2, -4 , and 7. For $s(x) = r(3x + 1)$, a zero occurs when $3x + 1$ equals one of those values.
- 2 The new zeros are $\frac{1}{3}$, $-\frac{5}{3}$, and 2. Their product is $\frac{1}{3} \times (-\frac{5}{3}) \times 2 = -\frac{10}{9}$.

Reasoning check The horizontal input transformation changes the product to $-\frac{10}{9}$.

Question 3

Math • Module 2 • Diagnostic Test 6

Problem-Solving and Data Analysis

Multiple choice

A 150-liter mixture is 42% acid by volume. It is made by combining solution A, which is 30% acid, and solution B, which is 60% acid. Let a be the liters of pure acid contributed by solution A and b the liters of pure acid contributed by solution B.

If a is $p\%$ less than b , what is p ?

A $33\frac{1}{3}$

B 40

C 20

D 25

Math • Module 2 • Question 3

Worked solution

Answer D

MIXTURES AND PERCENT COMPARISON

- 1 The 150-liter mixture contains $150(0.42) = 63$ liters of acid. Solution A contributes $90(0.30) = 27$ liters, so B contributes $63 - 27 = 36$ liters.
- 2 A is $36 - 27 = 9$ liters less than B. Relative to B, that is $9 \div 36 = 25\%$.

Reasoning check Solution A contains 25% less acid than solution B.

Question 4

Math · Module 2 · Diagnostic Test 6

Geometry and Trigonometry

Multiple choice

A circle has center $(4, -3)$ and radius 5. The point $(7, 1)$ lies on the circle. A line is tangent to the circle at $(7, 1)$.

What is the y -intercept of the tangent line?

A $\frac{31}{4}$

B 7

C $\frac{21}{4}$

D $\frac{25}{4}$

Math · Module 2 · Question 4

Worked solution

Answer D

TANGENT LINES

1 The radius from the center $(4, -3)$ to $(7, 1)$ has slope $(1 - (-3)) \div (7 - 4) = \frac{4}{3}$.

2 A tangent is perpendicular, so its slope is $-\frac{3}{4}$. Through $(7, 1)$, its y -intercept is $1 + (\frac{3}{4})(7) = \frac{25}{4}$.

Reasoning check The tangent line has y -intercept $\frac{25}{4}$.

Question 5

Math · Module 2 · Diagnostic Test 6

Algebra

Multiple choice

$$7x - 2y = 4y - 3$$

$$ay = bx + 9$$

In the system, a and b are constants. If the system has infinitely many solutions,

what is the value of $a + b$?

A 18

B -39

C 21

D 39

Math · Module 2 · Question 5

Worked solution

Answer D

INFINITELY MANY SOLUTIONS

- 1 Rewrite the first equation $7x - 2y = 4y - 3$ as $7x - 6y = -3$. Rewrite $ay = bx + 9$ as $bx - ay = -9$.
- 2 For infinitely many solutions, the second equation must be 3 times the first: $b = 21$ and $a = 18$. Thus $a + b = 39$.

Reasoning check The second equation is exactly a scalar multiple of the first when $a + b = 39$.

Question 6

Math · Module 2 · Diagnostic Test 6

Advanced Math

Multiple choice

For $k \neq 3$, the equation

$$(k - 3)x^2 + 6x + (k + 1) = 0$$

has exactly one real solution.

What is the sum of all possible values of k ?

A 6

B 0

C 2

D -2

Math · Module 2 · Question 6

Worked solution

Answer **C**

DISCRIMINANT

- 1 Because $k \neq 3$, the equation remains quadratic. Exactly one real root requires discriminant zero: $36 - 4(k - 3)(k + 1) = 0$.
- 2 Thus $(k - 3)(k + 1) = 9$, or $k^2 - 2k - 12 = 0$. The two possible values have sum 2.

Reasoning check The sum of the possible values is 2.

Question 7

Math · Module 2 · Diagnostic Test 6

Algebra

Multiple choice

The lines $3x + 2y = 12$ and $x - y = -1$ intersect at point P . Line m passes through P and is perpendicular to the line $4x - 3y = 10$.

What is the x -coordinate of the x -intercept of line m ?

A 5

B 6

C 8

D 4

Math · Module 2 · Question 7

Worked solution

Answer **B**

PERPENDICULAR LINES

1 The intersection of $3x + 2y = 12$ and $x - y = -1$ is $(2, 3)$.

2 The line $4x - 3y = 10$ has slope $\frac{4}{3}$, so a perpendicular through $(2, 3)$ has slope $-\frac{3}{4}$. Its equation gives an x -intercept of 6.

Reasoning check The perpendicular line crosses the x -axis at $x = 6$.

Question 8

Math · Module 2 · Diagnostic Test 6

Advanced Math

Multiple choice

For $x > 0$,

$$\frac{\sqrt[3]{x^{2m+1}}\sqrt{x^5}}{x^{m/2}} = x^{10/3}.$$

What is the value of m ?

A 2

B 3

C 1

D 5

Math · Module 2 · Question 8

Worked solution

Answer **B**

EXPONENT RULES

1 Use $\sqrt[3]{x^{2m+1}} = x^{(2m+1)/3}$ and $\sqrt{x^5} = x^{5/2}$.

2 The exponent equation is $(2m+1)/3 + 5/2 - m/2 = 10/3$. Multiplying by 6 gives $4m + 2 + 15 - 3m = 20$, so $m = 3$.

Reasoning check The exponent balance gives $m = 3$.

Question 9

Math • Module 2 • Diagnostic Test 6

Problem-Solving and Data Analysis

Multiple choice

A study records membership in a robotics club and enrollment in an honors course for 80 students.

	Honors	Not honors
Robotics club	28	12
Not in club	20	20

A student is chosen at random from those who are in the robotics club or enrolled in honors. What is the probability that the student is in both groups?

A $\frac{2}{3}$

B $\frac{7}{12}$

C $\frac{7}{20}$

D $\frac{7}{15}$

Math • Module 2 • Question 9

Worked solution

Answer D

TWO-WAY TABLE PROBABILITY

1 The number in both groups is 28. The union is $28 + 12 + 20 = 60$, counting the overlap once.

2 The requested sample is that union, so the probability is $28 \div 60 = \frac{7}{15}$.

Reasoning check Use the union as the denominator because the student is selected from the robotics-or-honors group.

Question 10

Math · Module 2 · Diagnostic Test 6

Geometry and Trigonometry

Multiple choice

Two cones are similar. The ratio of the lateral surface area of the larger cone to that of the smaller cone is $25 : 9$. The difference between their volumes is 784π cubic units.

What is the volume of the smaller cone?

A 144π

B 400π

C 288π

D 216π

Math · Module 2 · Question 10

Worked solution

Answer D

SIMILAR SOLIDS

1 Lateral area scales as the square of the linear scale factor. From $25:9$, the radius and height scale as $5:3$.

2 Volumes scale as $5^3:3^3 = 125:27$. If the smaller volume is V , then $(\frac{125}{27})V - V = 784\pi$, so $V = 216\pi$.

Reasoning check The smaller cone has volume 216π .

Question 11

Math · Module 2 · Diagnostic Test 6

Algebra

Multiple choice

A consultant charges a fixed service fee plus \$90 per hour for the first 4 hours of a job. For each hour beyond 4, the hourly charge is 25% lower. A 2-hour job costs \$240.

How much does a 10-hour job cost?

A \$870

B \$825

C \$780

D \$802.50

Math · Module 2 · Question 11

Worked solution

Answer **B**

PIECEWISE PRICING

1 A 2-hour job costs $F + 2(90) = 240$, so the fixed fee is $F = 60$.

2 The first 4 hours cost $60 + 4(90) = 420$. The remaining 6 hours cost $6(67.50) = 405$, for a total of 825.

Reasoning check The 10-hour job costs \$825.

Question 12

Math · Module 2 · Diagnostic Test 6

Advanced Math

Multiple choice

The function j is defined by $j(x) = 2(4)^x - 10$. A second function is defined by $g(x) = -2j(x - 1) + 6$.

What is the y -intercept of the graph of $y = g(x)$?

A (0, 21)

B (0, 19)

C (0, 25)

D (0, 29)

Math · Module 2 · Question 12

Worked solution

Answer C

EXPONENTIAL TRANSFORMATIONS

1 Evaluate $j(-1) = 2(4^{-1}) - 10 = \frac{1}{2} - 10 = -1\frac{1}{2}$.

2 Then $g(0) = -2j(-1) + 6 = -2(-1\frac{1}{2}) + 6 = 25$.

Reasoning check The y -intercept is 25.

Question 13

Math • Module 2 • Diagnostic Test 6

Problem-Solving and Data Analysis

Multiple choice

A linear model predicts a plant's height h , in centimeters, t days after planting by $h = 0.62t + 18.4$. For one plant observed at $t = 20$, the residual is -3.1 cm.

What is the plant's observed height?

A 30.8 cm

B 33.9 cm

C 37.0 cm

D 27.7 cm

Math • Module 2 • Question 13

Worked solution

Answer D

RESIDUALS

1 The predicted height at $t = 20$ is $0.62(20) + 18.4 = 30.8$.

2 Residual equals observed minus predicted. Thus $\text{observed} - 30.8 = -3.1$, so the observed height is 27.7.

Reasoning check The plant's observed height is 27.7 cm.

Question 14

Math · Module 2 · Diagnostic Test 6

Algebra

Multiple choice

A workshop produces tables x and chairs y . The constraints are

$$2x + y \leq 50, \quad x + y \geq 30, \quad y \geq 10.$$

Both x and y are nonnegative integers.

What is the greatest possible value of x ?

A 20

B 25

C 18

D 22

Math · Module 2 · Question 14

Worked solution

Answer **A**

LINEAR INEQUALITIES

- 1 The constraints require $y \geq 10$, $x + y \geq 30$, and $2x + y \leq 50$.
- 2 To maximize x , take the smallest allowable $y = 10$. Then $2x + 10 \leq 50$, so $x \leq 20$, and $x = 20$ is feasible.

Reasoning check The greatest possible value of x is 20.

Question 15

Math · Module 2 · Diagnostic Test 6

Advanced Math

Multiple choice

The polynomial $p(x) = x^3 + ax^2 + bx + 24$ has zeros -2 , 3 , and a third positive zero.

What is the value of $a - b$?

A -7

B 3

C -3

D 7

Math · Module 2 · Question 15

Worked solution

Answer C

POLYNOMIAL ROOTS

- 1 The roots are -2 , 3 , and t . Since the constant term is 24 , $-(-2)(3)t = 24$, so $t = 4$.
- 2 The sum of roots is 5 , so $-a = 5$ and $a = -5$. The pairwise sum is $b = -2$, hence $a - b = -3$.

Reasoning check The third root is 4 , and the coefficient relations give $a - b = -3$.

Question 16

Math · Module 2 · Diagnostic Test 6

Geometry and Trigonometry

Multiple choice

The equation

$$x^2 + y^2 + ax - 8y + 4 = 0$$

represents a circle of radius 5, where $a > 0$. A square is circumscribed about the circle so that each side of the square is tangent to the circle.

What is the value of a^2 plus the area of the square?

A 152

B 169

C 125

D 144

Math · Module 2 · Question 16

Worked solution

Answer **A**

CIRCLE AND SQUARE GEOMETRY

- 1 Completing the square in $x^2 + y^2 + ax - 8y + 4 = 0$ shows $r^2 = a^2/4 + 12 = 25$, so $a^2 = 52$.
- 2 The circumscribed square has side $2r = 10$ and area 100. The requested sum is $52 + 100 = 152$.

Reasoning check The sum of a^2 and the square's area is 152.

Question 17

Math • Module 2 • Diagnostic Test 6

Algebra

SPR

$$\frac{3a}{8}x = \frac{3}{2} + \frac{15}{8}y$$
$$\frac{18}{7}y - 4 = \frac{4b}{7}x - 9$$

The positive constants a and b make the system have no solution. If b is $p\%$ less than a ,

what is the value of p ?

• Student-produced response • enter a numerical answer

Math • Module 2 • Question 17

Worked solution

Answer 10

PARALLEL SYSTEMS AND PERCENT CHANGE

- 1 The first equation simplifies to $ax - 5y = 4$. The second simplifies to $4bx - 18y = 35$.
- 2 For parallel lines, the coefficient ratios match: $4b \div a = 18 \div 5$, so $b = 0.9a$. Thus b is 10% less than a , and $p = 10$.

Reasoning check The coefficient comparison gives a 10% decrease from a to b .

Question 18

Math • Module 2 • Diagnostic Test 6

Advanced Math

SPR

The parabola $y = x^2 - 8x + c$ and the line $y = 2x - 5$ intersect at two points. The x -coordinates of the intersection points differ by 4.

What is the value of c ?

• Student-produced response • enter a numerical answer

Math • Module 2 • Question 18

Worked solution

Answer 16

QUADRATIC ROOTS

- 1 At an intersection, $x^2 - 8x + c = 2x - 5$, so $x^2 - 10x + (c + 5) = 0$.
- 2 The roots differ by 4, so $16 = 10^2 - 4(c + 5)$. Solving gives $c = 16$.

Reasoning check The intersection condition produces $c = 16$.

Question 19

Math • Module 2 • Diagnostic Test 6

Problem-Solving and Data Analysis

SPR

A theater can seat at most 240 people. Adult tickets cost \$15 and child tickets cost \$9. At least 35% of all tickets sold must be child tickets, and ticket revenue must be at least \$3,000.

What is the least possible number of adult tickets that can be sold?

☒ **Student-produced response • enter a numerical answer**

Math • Module 2 • Question 19

Worked solution

Answer 140

LINEAR PROGRAMMING

- 1 Let A be adult tickets and C be child tickets. The constraints include $A + C \leq 240$, $C \geq 0.35(A + C)$, and $15A + 9C \geq 3000$.
- 2 For a fixed A , the largest possible C is $240 - A$, so revenue is at most $15A + 9(240 - A) = 6A + 2160$. Requiring at least 3000 gives $A \geq 140$.
- 3 The point $A = 140$, $C = 100$ meets every constraint, so the minimum is 140 adults.

Reasoning check The least feasible number of adult tickets is 140.

Question 20

Math · Module 2 · Diagnostic Test 6

Algebra SPR

The table shows three values of x and their corresponding values of y for a linear relationship. The x -intercept is -4 . The constant d is negative.

x	y
-7	$d + 3$
-4	$4k + 6d$
$1 - 3d$	8

What is the value of dk ?

Student-produced response · enter a numerical answer

Math · Module 2 · Question 20

Worked solution

Answer **-169/6**

LINEAR RELATIONSHIPS AND TABLES

- Because the x -intercept is -4 , the y -value at $x = -4$ is zero: $4k + 6d = 0$, so $k = -3d/2$.
- The slope from $(-7, d + 3)$ to $(-4, 0)$ is $-(d + 3)/3$. The slope from $(-4, 0)$ to $(1 - 3d, 8)$ is $8/(5 - 3d)$.
- Equating slopes gives $3d^2 + 4d - 39 = 0$, so $d = 3$ or $d = -13/3$. Since d is negative, $d = -13/3$ and $k = 13/2$. Thus $dk = -169/6$.

Reasoning check The accepted numerical equivalents are -16% , -28.1667 , and -28.167 .

Question 21

Math · Module 2 · Diagnostic Test 6

Advanced Math

SPR

The function $f(x) = ax^2 + bx + c$ satisfies $f(-11) = f(-1)$. The coefficient b is an integer less than -3 .

What is the greatest possible value of $a + b$?

 Student-produced response • enter a numerical answer

Math · Module 2 · Question 21

Worked solution

Answer **-13/3**

POLYNOMIAL SYMMETRY

- 1 For $f(x) = ax^2 + bx + c$, the condition $f(-11) = f(-1)$ gives $120a - 10b = 0$, hence $b = 12a$.
- 2 Therefore $a + b = 13b/12$. Since b is an integer less than -3 , its greatest possible value is $b = -4$, giving $a + b = -1\frac{1}{3}$.

Reasoning check The largest possible value is $-1\frac{1}{3}$; equivalent accepted decimals are -4.3333 and -4.333 .

Question 22

Math · Module 2 · Diagnostic Test 6

Geometry and Trigonometry

SPR

A square has diagonal length $16\sqrt{2}$. A circle is inscribed in the square, and a regular hexagon is inscribed in that circle.

What is the perimeter of the regular hexagon?

 Student-produced response • enter a numerical answer

Math • Module 2 • Question 22

Worked solution

Answer 48

INSCRIBED FIGURES

- 1 A square with diagonal $16\sqrt{2}$ has side 16, because its diagonal is $s\sqrt{2}$.
- 2 The circle's radius is half the diagonal, 8. An inscribed regular hexagon has side length equal to the circle's radius, so its perimeter is $6(8) = 48$.

Reasoning check The regular hexagon's perimeter is 48.

PART II

Answer key

Select any answer chip or explanation link to jump to that item's worked solution.



Q	ANSWER	SOLUTION
1	C	Explain ↗
2	C	Explain ↗
3	C	Explain ↗
4	B	Explain ↗
5	B	Explain ↗
6	B	Explain ↗
7	C	Explain ↗
8	A	Explain ↗
9	B	Explain ↗
10	B	Explain ↗
11	B	Explain ↗
12	D	Explain ↗
13	D	Explain ↗
14	D	Explain ↗
15	B	Explain ↗
16	C	Explain ↗
17	D	Explain ↗
18	D	Explain ↗
19	D	Explain ↗
20	B	Explain ↗
21	B	Explain ↗
22	A	Explain ↗

Q	ANSWER	SOLUTION
23	A	Explain ↗
24	D	Explain ↗
25	D	Explain ↗
26	A	Explain ↗
27	C	Explain ↗





Q	ANSWER	SOLUTION
1	C	Explain ↗
2	A	Explain ↗
3	D	Explain ↗
4	C	Explain ↗
5	D	Explain ↗
6	C	Explain ↗
7	B	Explain ↗
8	B	Explain ↗
9	B	Explain ↗
10	A	Explain ↗
11	C	Explain ↗
12	C	Explain ↗
13	A	Explain ↗
14	D	Explain ↗
15	B	Explain ↗
16	C	Explain ↗
17	D	Explain ↗
18	D	Explain ↗
19	D	Explain ↗
20	B	Explain ↗
21	B	Explain ↗
22	B	Explain ↗

Q	ANSWER	SOLUTION
23	A	Explain ↗
24	C	Explain ↗
25	D	Explain ↗
26	D	Explain ↗
27	A	Explain ↗



Q	ANSWER	SOLUTION
1	C	Explain ↗
2	A	Explain ↗
3	A	Explain ↗
4	B	Explain ↗
5	B	Explain ↗
6	B	Explain ↗
7	C	Explain ↗
8	B	Explain ↗
9	D	Explain ↗
10	A	Explain ↗
11	B	Explain ↗
12	C	Explain ↗
13	B	Explain ↗
14	C	Explain ↗
15	B	Explain ↗
16	C	Explain ↗
17	C	Explain ↗
18	24	Explain ↗
19	21	Explain ↗
20	75	Explain ↗
21	9	Explain ↗
22	2	Explain ↗

Q	ANSWER	SOLUTION
1	A	Explain ↗
2	B	Explain ↗
3	D	Explain ↗
4	D	Explain ↗
5	D	Explain ↗
6	C	Explain ↗
7	B	Explain ↗
8	B	Explain ↗
9	D	Explain ↗
10	D	Explain ↗
11	B	Explain ↗
12	C	Explain ↗
13	D	Explain ↗
14	A	Explain ↗
15	C	Explain ↗
16	A	Explain ↗
17	10	Explain ↗
18	16	Explain ↗
19	140	Explain ↗
20	-169/6	Explain ↗
21	-13/3	Explain ↗
22	48	Explain ↗